

Equilibrium of Forces

Translational & rotational equilibrium · resolving forces · cables & ropes · beams, levers & seesaws

Physics — Grade 11
Explanation &
Questions
 $g = 10 \text{ m/s}^2$

1 What does equilibrium mean?

A body is in **equilibrium** when the forces acting on it produce **no change** in its state of motion: it does not start moving (or keeps moving with a constant velocity), and it does not start rotating.

Type	What the body does	Condition
Translational equilibrium	at rest, or moving in a straight line with a constant velocity	$\Sigma F = 0$
Rotational equilibrium	not rotating, or rotating with a constant angular speed	$\Sigma M = 0$
Static (complete) equilibrium	completely at rest	$\Sigma F = 0$ and $\Sigma M = 0$

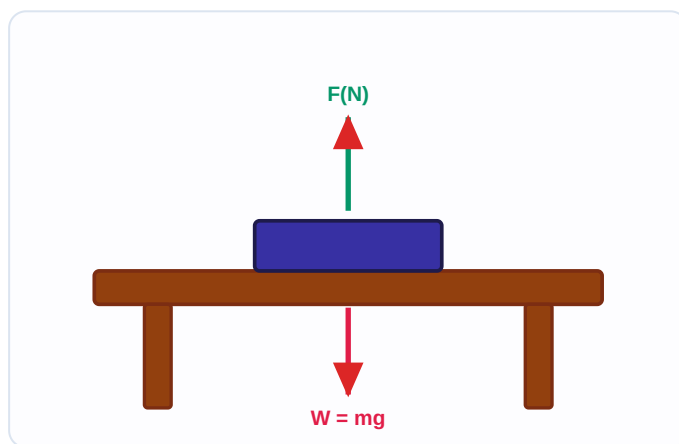


Fig. 1 — a book resting on a table: the weight is balanced by the normal force.

Remember

Equilibrium does **not** mean “no forces”. Many forces may act on the body, but their **resultant** is zero and their **total moment** is zero.

2 The first condition — the resultant force is zero

For translational equilibrium the vector sum of all the forces must vanish. In practice we resolve every force into a horizontal (x) and a vertical (y) component and apply the condition to each direction separately:

$$\Sigma F_x = 0 \quad \text{and} \quad \Sigma F_y = 0$$

forces to the right = forces to the left | forces upwards = forces downwards

A force F that makes an angle θ with the horizontal has the components

$$F_x = F \cos \theta \quad F_y = F \sin \theta$$

θ is measured from the **horizontal**

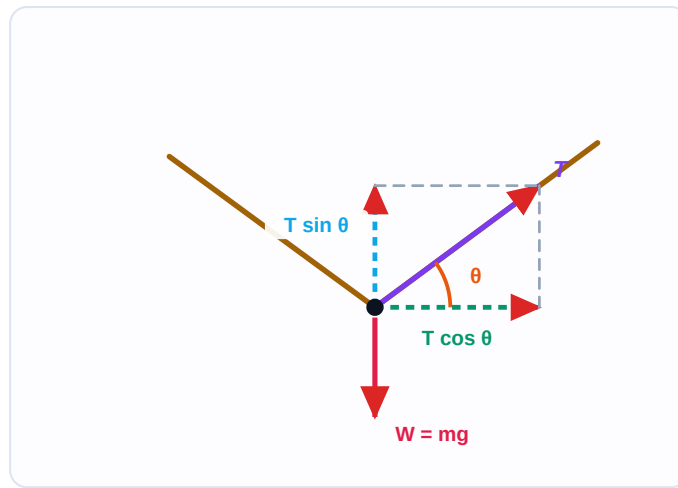


Fig. 2 — a tension T resolved into a horizontal and a vertical component.

Three forces at one point

If three forces keep a point in equilibrium, then the resultant of **any two** of them is equal in magnitude and opposite in direction to the third one.

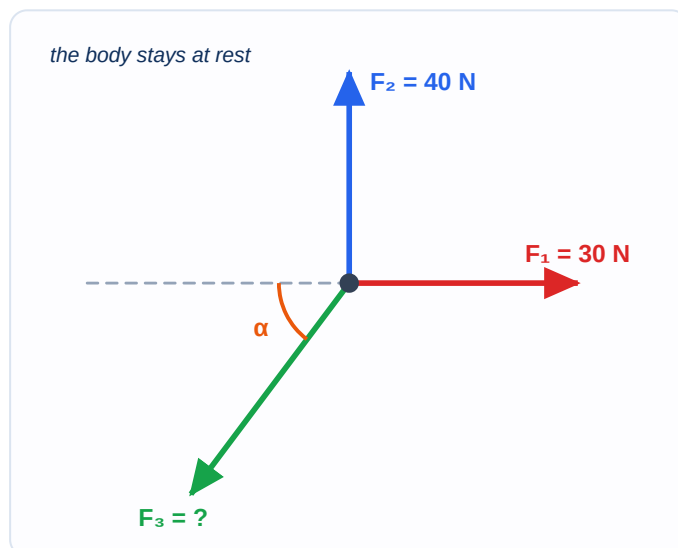


Fig. 3 — three forces acting at one point

3 A weight hanging from two cables

A lamp of weight W hangs from the middle of two identical cables, each making an angle θ with the **horizontal** (Fig. 4). By symmetry the two tensions are equal. The horizontal components cancel, and the two vertical components together carry the weight:

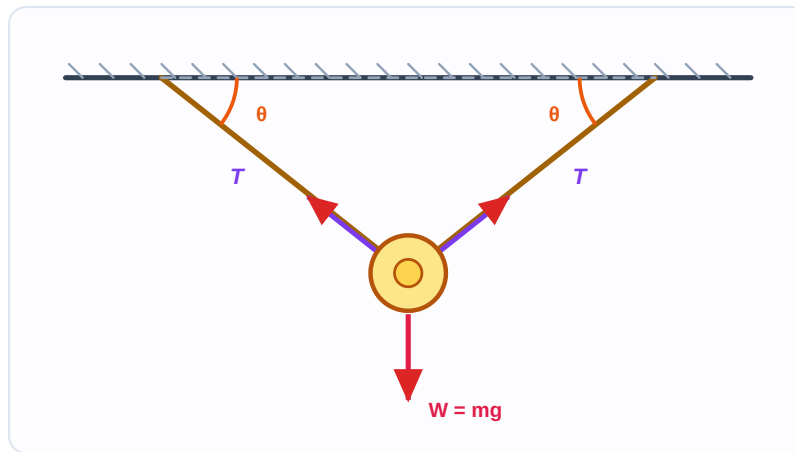


Fig. 4 — a lamp supported by two symmetric cables.

$$2 T \sin \theta = W \Rightarrow T = W / (2 \sin \theta)$$

θ = angle between each cable and the horizontal

θ (with the horizontal)	90°	60°	30°	10°	5°	1°
T / W	0.5	0.58	1	2.9	5.7	28.6

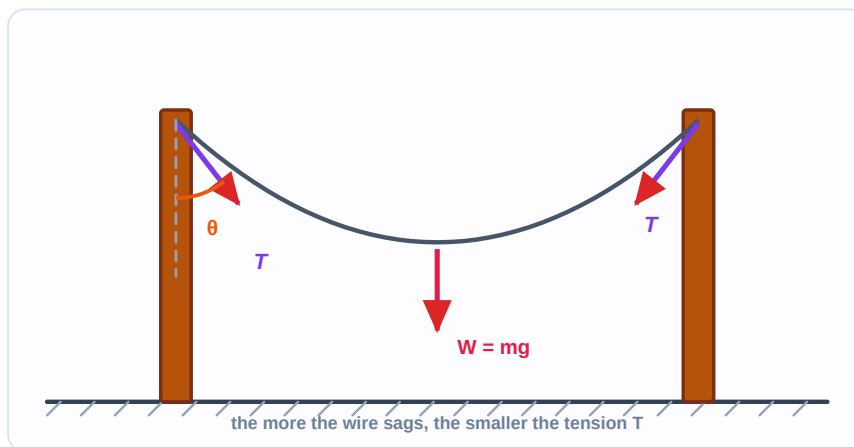


Fig. 5 — the flatter the wire, the larger the tension.

Why can a stretched wire never be perfectly straight?

As $\theta \rightarrow 0$, $\sin \theta \rightarrow 0$ and $T = W / (2 \sin \theta)$ becomes enormous. A wire carrying any weight must therefore sag a little, however strongly it is pulled.

4 A lamp pulled aside by a horizontal rope

A lamp hangs from a cable fixed to the ceiling and is pulled aside by a horizontal rope, so that the cable makes an angle θ with the **vertical** (Fig. 6). Three forces act on the lamp: the tension T , the horizontal pull P and the weight W .

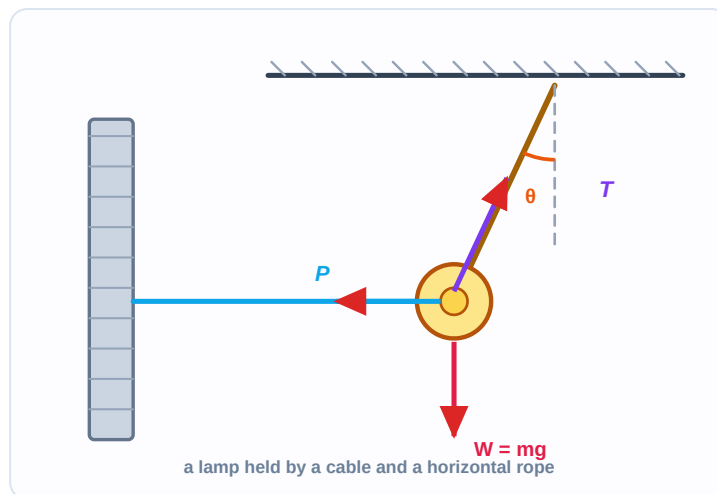


Fig. 6 — three forces keep the lamp in equilibrium.

$$\text{vertical : } T \cos \theta = W \quad \text{horizontal : } T \sin \theta = P$$

θ = angle between the cable and the **vertical** | dividing : $\tan \theta = P / W$

5 The second condition — the total moment is zero

A body can have zero resultant force and still **rotate** (two equal and opposite forces that do not act along the same line). For rotational equilibrium the total moment about **any** point must be zero:

$$\Sigma M = 0 \Rightarrow \text{sum of clockwise moments} = \text{sum of anticlockwise moments}$$

the moments are taken about the same point

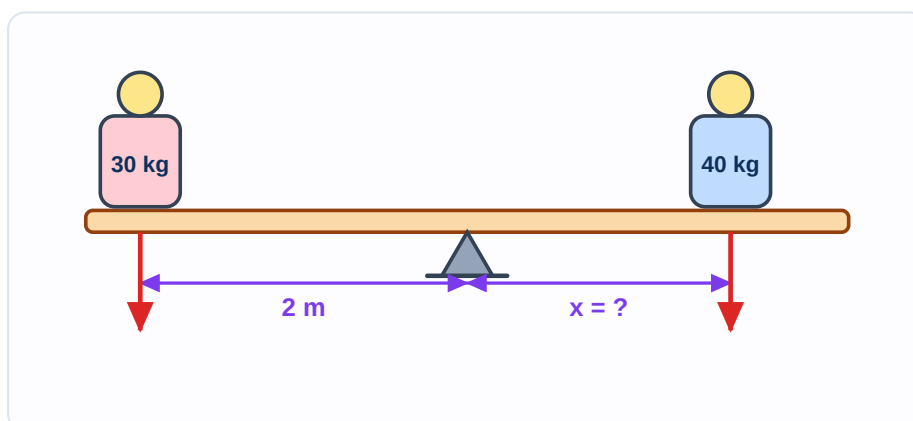


Fig. 7 — a balanced seesaw: the two moments about the pivot are equal and opposite.

Choosing the point

You may take moments about **any** point. Choose the point where an **unknown force** acts: its moment about that point is zero, so it disappears from the equation.

The weight of a uniform body

The weight of a **uniform** rod, beam or plank acts at its **middle point** (its centre of gravity).

6 Using both conditions — a beam on two supports

A uniform beam rests on two supports A and B and carries a load. The supports push the beam upwards with the reactions R_1 and R_2 (Fig. 8). Two equations give the two unknowns:

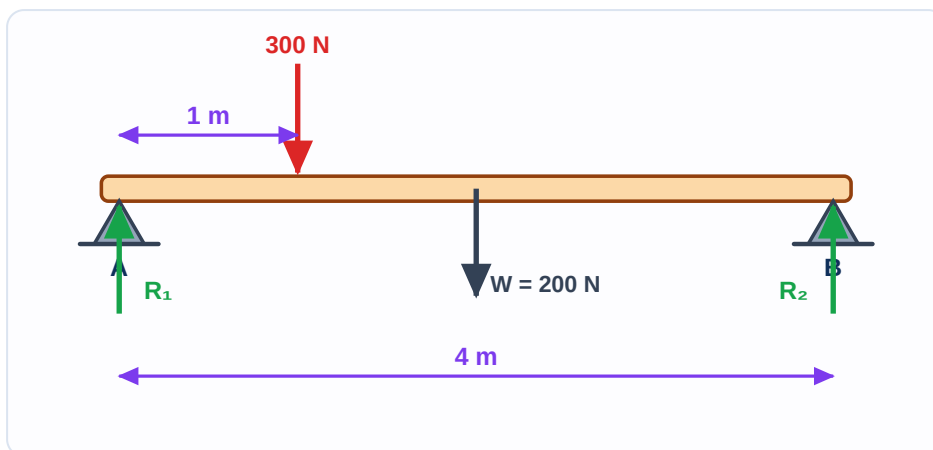


Fig. 8 — a beam in complete equilibrium.

1. $\Sigma F_y = 0$: $R_1 + R_2 = \text{load} + \text{weight of the beam}$
2. $\Sigma M = 0$ about A : $R_2 \times AB = (\text{load} \times \text{its distance from A}) + (W \times \text{half the length})$

7 How to solve an equilibrium problem

1. Draw the body alone and mark **every** force acting on it (weight, tensions, reactions, pushes, friction).
2. Resolve the inclined forces into horizontal and vertical components.
3. Write $\Sigma F_x = 0$ and $\Sigma F_y = 0$.
4. If the body can rotate, take moments about a point through which an unknown force passes and write $\Sigma M = 0$.
5. Solve, and check the answer with the equation you did not use.

★ Key facts you must memorise

- Translational equilibrium : $\Sigma \mathbf{F} = \mathbf{0}$ (at rest or constant velocity).
- Rotational equilibrium : $\Sigma \mathbf{M} = \mathbf{0}$ (clockwise moments = anticlockwise moments).
- Components : $F_x = F \cos \theta$, $F_y = F \sin \theta$ (θ from the horizontal).
- Two symmetric cables : $T = W / (2 \sin \theta)$; the flatter the cables, the larger the tension.
- Three forces in equilibrium : any two have a resultant equal and opposite to the third.
- The weight of a uniform body acts at its middle.

Common mistakes

- Forgetting the **weight of the beam or rod** itself.
- Using sin instead of cos: check whether θ is measured from the horizontal or from the vertical.
- Writing $T = W$ for a cable that is not vertical.
- Taking moments about different points in the same equation.
- Forgetting to change grams to kilograms and centimetres to metres.

8 Solved examples

Example 1 — a lamp on two cables

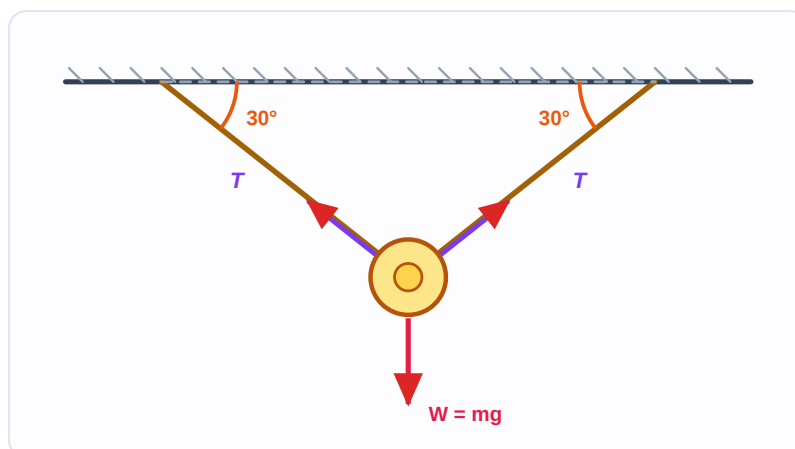


Fig. 9 — each cable makes 30° with the horizontal.

Given: A lamp of mass 0.6 kg hangs from two identical cables, each making 30° with the horizontal. Find the tension in each cable.

Solution

$$W = m g = 0.6 \times 10 = 6 \text{ N}$$

the horizontal components cancel ; vertically : $2 T \sin 30^\circ = W$

$$2 T \times 0.5 = 6 \quad \Rightarrow \quad T = 6 \text{ N}$$

Notice that each cable carries a tension **equal to the whole weight** at 30° .

$$T = 6 \text{ N}$$

Example 2 — a lamp pulled aside by a rope

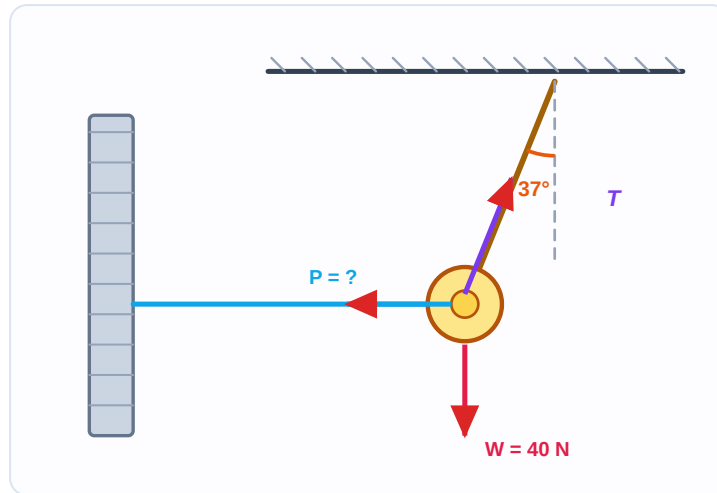


Fig. 10 — the cable makes 37° with the vertical.

Given: A lamp of weight 40 N is pulled aside by a horizontal rope until its cable makes 37° with the vertical. Find the tension T in the cable and the pull P. ($\sin 37^\circ = 0.6$, $\cos 37^\circ = 0.8$)

Solution

$$\text{vertical : } T \cos 37^\circ = W \Rightarrow T = 40 / 0.8 = 50 \text{ N}$$

$$\text{horizontal : } P = T \sin 37^\circ = 50 \times 0.6 = 30 \text{ N}$$

Check: $\tan 37^\circ = P / W = 30 / 40 = 0.75$ ✓

$$T = 50 \text{ N}, P = 30 \text{ N}$$

Example 3 — the third force

Given: Two forces act on a point: 30 N towards the east and 40 N towards the north (Fig. 3). Find the third force that keeps the point in equilibrium.

Solution

$$\text{resultant of the two forces : } R = \sqrt{30^2 + 40^2} = \sqrt{2500} = 50 \text{ N}$$

$$\text{direction of R : } \tan \alpha = 40 / 30 \Rightarrow \alpha = 53^\circ \text{ north of east}$$

the third force is equal and opposite to R

$$F_3 = 50 \text{ N, at } 53^\circ \text{ south of west}$$

Example 4 — a box pulled at an angle with a constant velocity

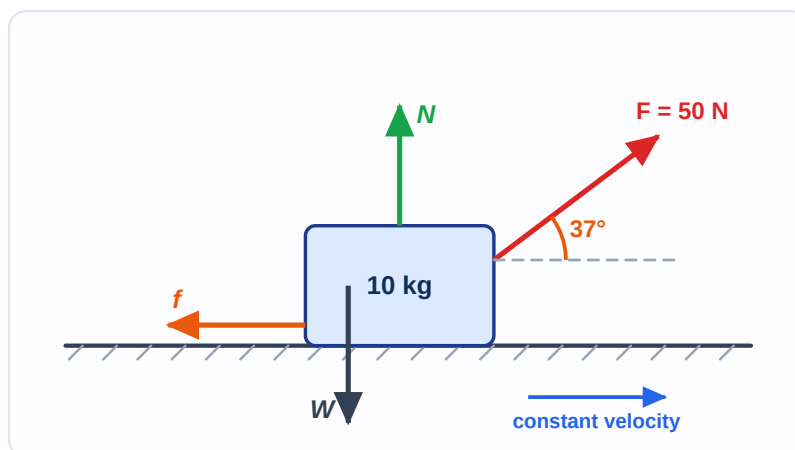


Fig. 11 — the box moves with a constant velocity.

Given: A box of mass 10 kg is pulled along a horizontal floor by a force of 50 N inclined at 37° above the horizontal. The box moves with a **constant velocity**. Find the friction force and the normal force.

Solution

Constant velocity $\Rightarrow$ translational equilibrium $\Rightarrow \Sigma F_x = 0$ and $\Sigma F_y = 0$.

$$\text{horizontal : } f = F \cos 37^\circ = 50 \times 0.8 = 40 \text{ N}$$

$$\text{vertical : } N + F \sin 37^\circ = W \quad \Rightarrow \quad N = 100 - 50 \times 0.6 = 70 \text{ N}$$

The upward component of the pull makes the normal force **less** than the weight.

$$f = 40 \text{ N}, N = 70 \text{ N}$$

Example 5 — a balanced seesaw

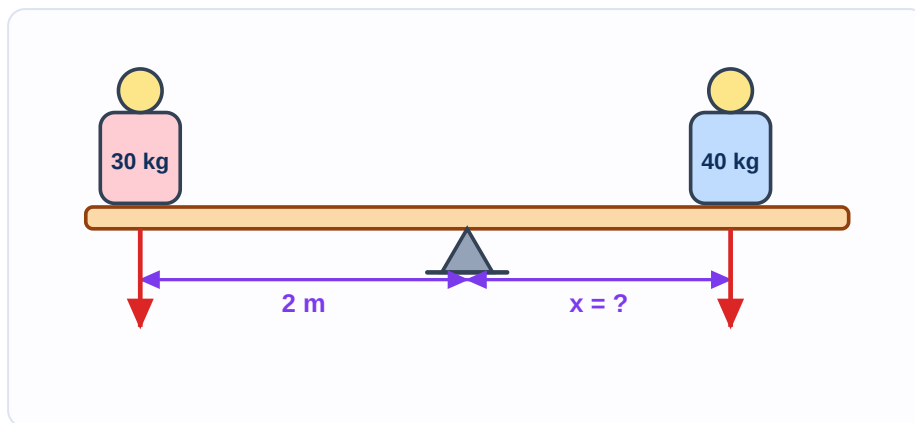


Fig. 12 — where must the heavier child sit?

Given: A 30 kg child sits 2 m from the pivot of a uniform seesaw pivoted at its middle. Where must a 40 kg child sit, on the other side, to balance it?

Solution

about the pivot : clockwise moments = anticlockwise moments
 $(30 \times 10) \times 2 = (40 \times 10) \times x$
 $600 = 400x \Rightarrow x = 1.5 \text{ m}$

The heavier child must sit **nearer** to the pivot.

x = 1.5 m from the pivot

Example 6 — the reactions on a beam

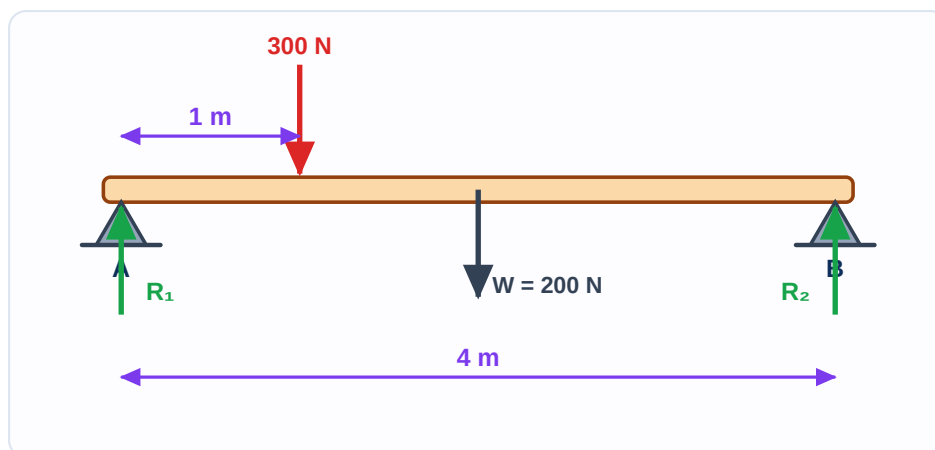


Fig. 13 — a uniform beam on two supports at its ends.

Given: A uniform beam of length 4 m and weight 200 N rests on supports at its ends A and B. A load of 300 N is placed 1 m from A. Find the reactions R_1 (at A) and R_2 (at B).

Solution

moments about A (R_1 disappears) :

$$R_2 \times 4 = (300 \times 1) + (200 \times 2) = 700 \Rightarrow R_2 = 175 \text{ N}$$

$$\text{vertical forces : } R_1 + R_2 = 300 + 200 = 500 \Rightarrow R_1 = 325 \text{ N}$$

Check with moments about B: $R_1 \times 4 = 300 \times 3 + 200 \times 2 = 1300 \Rightarrow R_1 = 325 \text{ N} \checkmark$

$$R_1 = 325 \text{ N} , R_2 = 175 \text{ N}$$

Example 7 — finding the weight of a metre rule

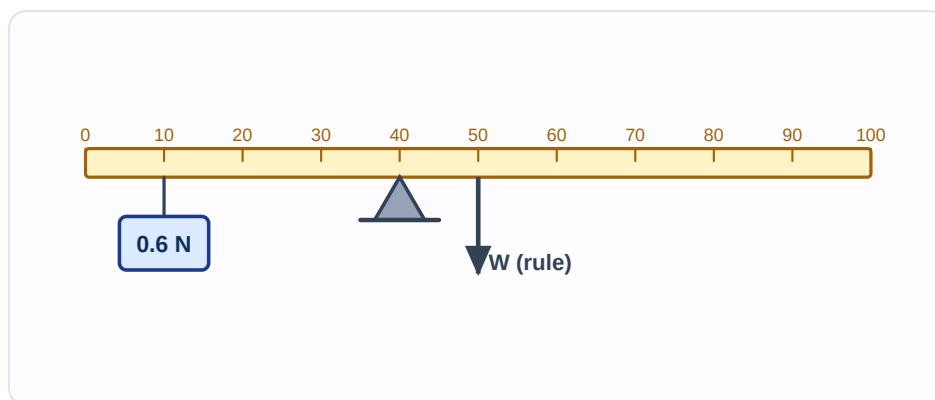


Fig. 14 — a uniform metre rule balanced on a knife-edge at the 40 cm mark.

Given: A uniform metre rule balances horizontally on a knife-edge at the 40 cm mark when a weight of 0.6 N hangs at the 10 cm mark. Find the weight of the rule.

Solution

the weight of the rule acts at its middle (50 cm), 10 cm = 0.1 m to the right of the pivot

the 0.6 N weight is 40 - 10 = 30 cm = 0.3 m to the left

moments about the pivot : $0.6 \times 0.3 = W \times 0.1 \Rightarrow W = 1.8 \text{ N}$

$$W = 1.8 \text{ N}$$

9 Exercises

20 questions. Take $g = 10 \text{ m/s}^2$, $\sin 37^\circ = 0.6$, $\cos 37^\circ = 0.8$, $\sin 30^\circ = 0.5$, $\cos 30^\circ = 0.866$, $\sin 45^\circ = 0.707$. Show every step and give every answer with its unit.

Q1 A body is in **complete (static)** equilibrium when:

MCQ

- ☐ A) the resultant force on it is zero only
- ☐ B) the total moment on it is zero only
- ☐ C) both the resultant force and the total moment are zero
- ☐ D) no forces act on it

Q2 A car moves along a straight road with a **constant velocity**. The resultant force acting on it is:

MCQ

- ☐ A) in the direction of motion
- ☐ B) zero
- ☐ C) opposite to the motion
- ☐ D) equal to its weight

Q3 Two forces of 8 N and 6 N act at right angles on a point. The magnitude of the third force that keeps the point in equilibrium is:

MCQ

- ☐ A) 2 N
- ☐ B) 14 N
- ☐ C) 10 N
- ☐ D) 48 N

Q4 A lamp of mass 2 kg hangs from two identical cables, each making 30° with the horizontal. The tension in each cable is:

MCQ

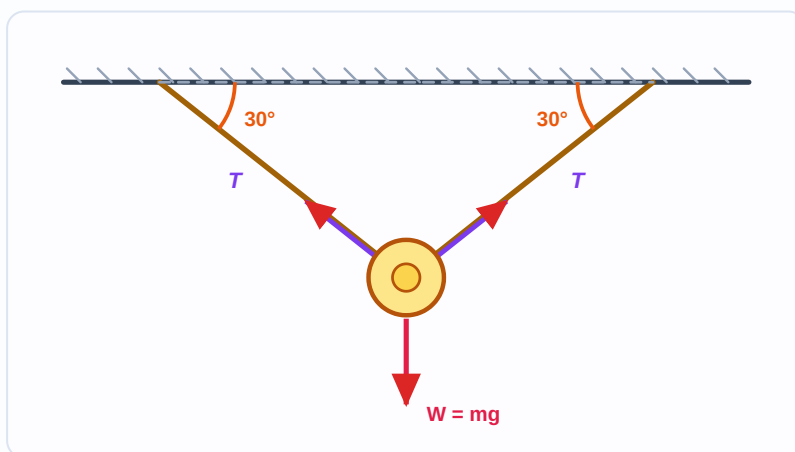


Fig. 15

- ☐ A) 10 N
- ☐ B) 40 N
- ☐ C) 11.5 N
- ☐ D) 20 N

Q5 For the lamp of Q4, if the two cables are made **flatter** (the angle with the horizontal decreases), the tension in each cable:

MCQ

A) increases

B) decreases

C) stays the same

D) becomes zero

Q6 A sign of mass 5 kg hangs from two identical cables, each making 60° with the horizontal. Find the tension in each cable. ($\sin 60^\circ = 0.866$)

Problem

Q7 A lamp of weight 24 N is pulled aside by a horizontal force P until its cable makes 37° with the vertical. Find the tension in the cable and the force P .

Problem

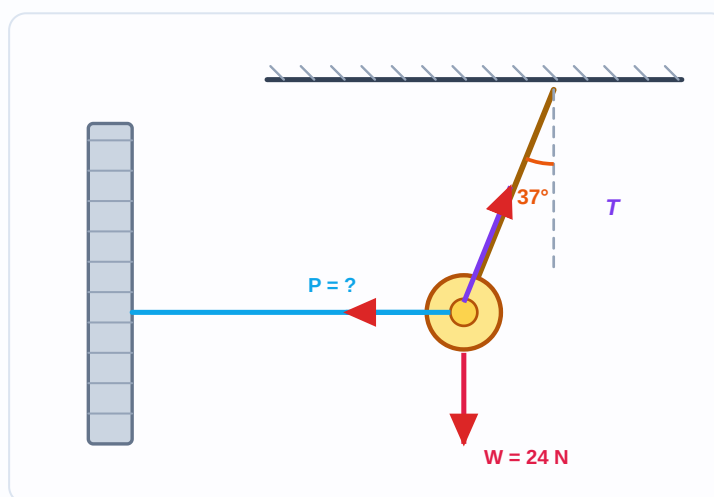


Fig. 16

Q8 A box is pulled along a horizontal floor by a horizontal force of 20 N and moves with a constant velocity. The friction force on the box is:

MCQ

A) zero

B) 20 N

C) more than 20 N

D) less than 20 N

Q9 A box of mass 8 kg is pulled along a horizontal floor by a force of 40 N at 30° above the horizontal, and moves with a constant velocity. Find the friction force and the normal force.

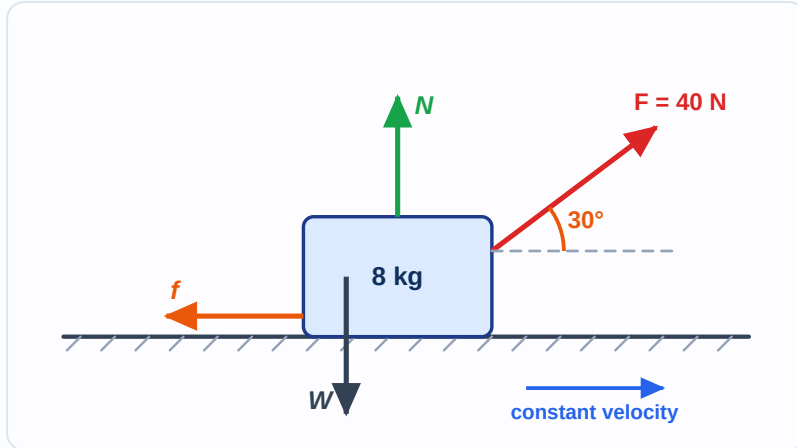


Fig. 17

Q10 A 25 kg child sits 2.4 m from the pivot of a uniform seesaw pivoted at its middle. A 30 kg child can balance it by sitting on the other side at a distance of: MCQ

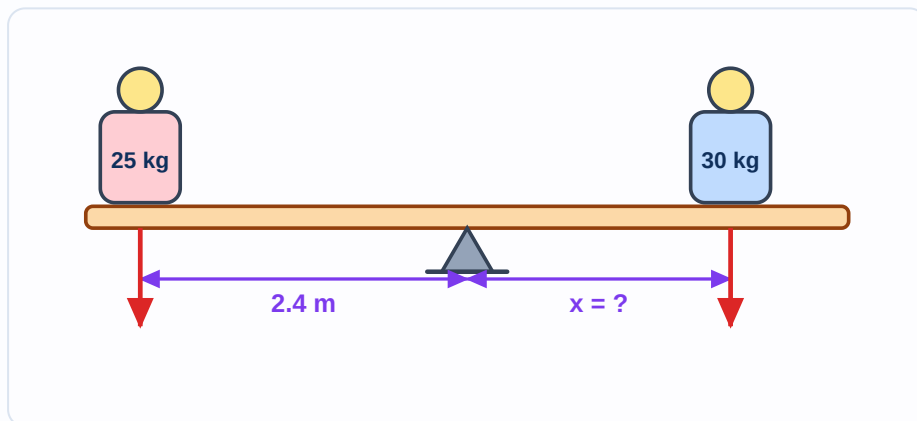


Fig. 18

A) 2 m

B) 2.88 m

C) 1.2 m

D) 2.4 m

Q11 A uniform metre rule of weight 10 N is pivoted at the 40 cm mark. The weight that must hang at the 0 cm mark to keep the rule horizontal is: MCQ

A) 10 N

B) 2.5 N

C) 4 N

D) 25 N

Q12 A uniform beam of length 6 m and weight 300 N rests on two supports at its ends A and B. A load of 600 N is placed 2 m from A. Find the two reactions.

Problem

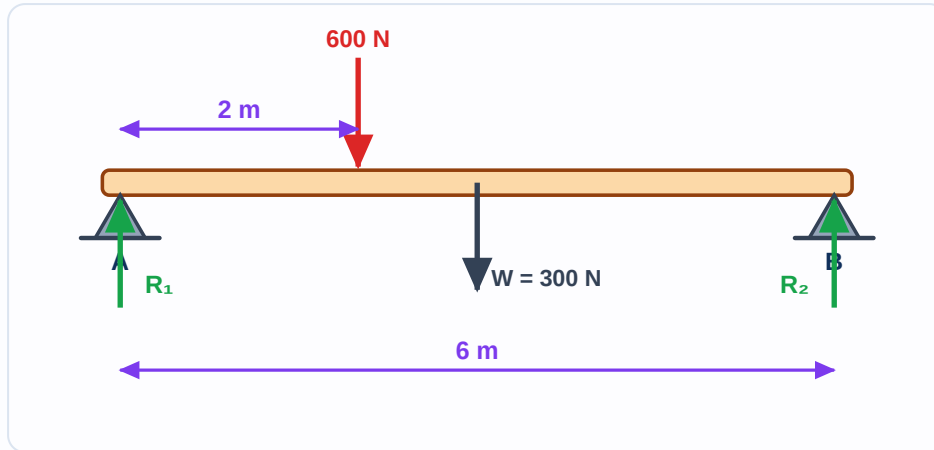


Fig. 19

Q13 A uniform beam rests on two supports at its ends and carries a load **exactly at its middle**. The two reactions are:

MCQ

A) zero

B) different

C) equal

D) each equal to the total weight

Q14 Two forces act on a point: 12 N towards the east and 5 N towards the south. Find the magnitude and the direction of the third force that keeps the point in equilibrium.

Problem

Q15 A picture of weight 14.1 N hangs from a nail by a string whose two halves each make 45° with the horizontal. The tension in the string is:

MCQ

A) 10 N

B) 14.1 N

C) 7.05 N

D) 20 N

Q16 A uniform metre rule of weight 1.5 N is pivoted at the 60 cm mark. At which mark must a weight of 1 N hang to keep the rule horizontal?

Problem

Q17 A uniform plank of length 4 m and weight 400 N rests on two supports: one at its end A and one 1 m from the other end B. The reaction of the second support is:

MCQ

A) 200 N

B) 133.3 N

C) 400 N

D) 266.7 N

Q18 A man of weight 700 N walks along a uniform plank of length 5 m and weight 300 N, supported at its end A and at C, 1 m from the other end B (Fig. 20). How far beyond C can the man walk before the plank starts to tip?

★ HOTS

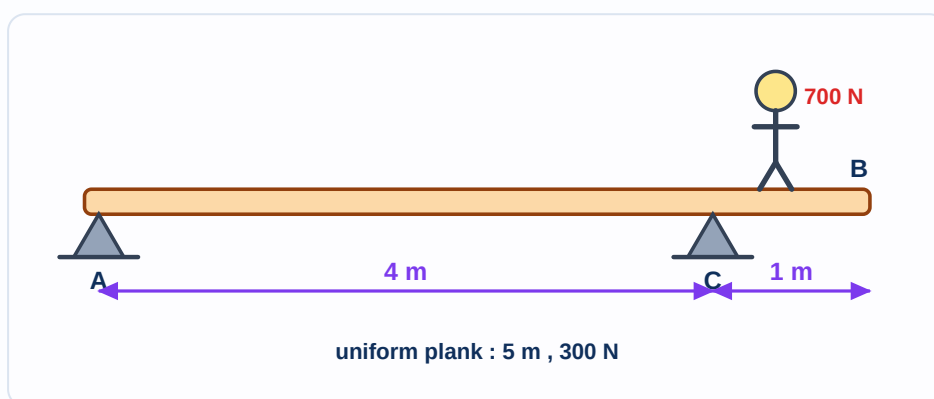


Fig. 20

Q19 When a body is in equilibrium under several forces, the total moment of these forces is zero:

MCQ

A) about its centre only

B) about any point

C) about the point of support only

D) only if all the forces are vertical

Q20 A uniform horizontal rod of length 2 m and weight 40 N is hinged to a wall at one end and ★ HOTS

held by a vertical rope at the other end. A load of 60 N hangs 0.5 m from the wall. Find the tension in the rope and the force of the hinge.

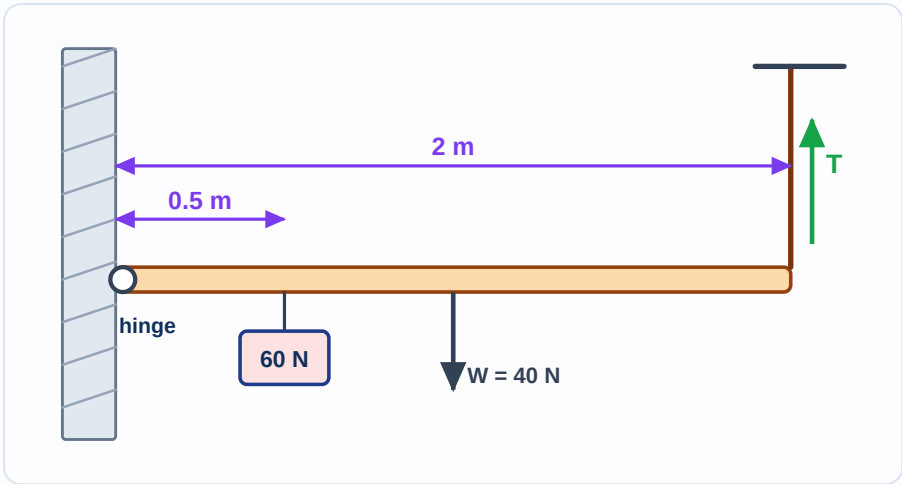


Fig. 21

10 Model answer — final answers only

Use this page to correct the exercises. Only the final answer is given, so the student must write every step of the working.

Q	Final answer	Q	Final answer
1	C) both the resultant force and the total moment are zero	2	B) zero
3	C) 10 N	4	D) 20 N
5	A) increases	6	T = 28.9 N
7	T = 30 N , P = 18 N	8	B) 20 N
9	f = 34.6 N , N = 60 N	10	A) 2 m
11	B) 2.5 N	12	R(A) = 550 N , R(B) = 350 N
13	C) equal	14	13 N , at 22.6° north of west
15	A) 10 N	16	at the 75 cm mark
17	D) 266.7 N	18	0.64 m beyond C
19	B) about any point	20	T = 35 N , hinge force = 65 N upwards